Name

Mathematics Advanced Year 11

Graphical Applications of the Derivative

Book 4

Version: 261021

Feedback:
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Syllabus Content

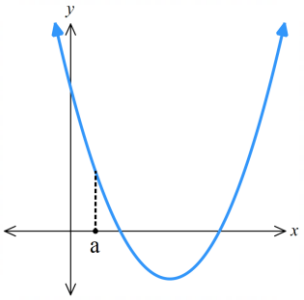
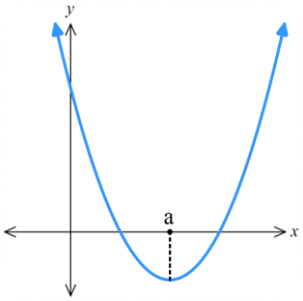
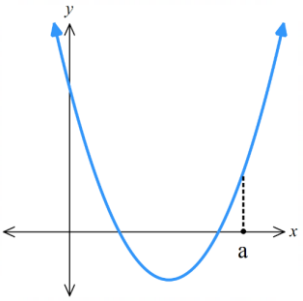
MAV-11-06 interprets the meaning of the derivative and determines the derivative of functions to solve problems

Graphical applications of the derivative

- Interpret $f(x)$ as increasing at $x = c$ when $f'(c) > 0$ and decreasing at $x = c$ when $f'(c) < 0$
- Describe the behaviour of a function at a point as stationary when the tangent at the point is parallel to the x-axis, and recognise that $f(x)$ is stationary at $x = c$ when $f'(c) = 0$
- Graph $y = f'(x)$ for a given graph of a function $y = f(x)$
- Numerically estimate the value of the derivative at a point on the graph of a power of x, with and without the use of digital tools
- Identify stationary points on the graph of a cubic function, and the values of x for which the function is increasing and/or decreasing, by first calculating the derivative, and justify conclusions

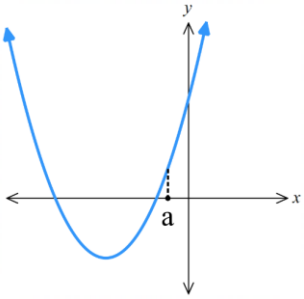
Increasing Decreasing Stationary Graphically

ⓘ Behaviour of a function at a point

Decreasing at $x = a$	Stationary at $x = a$	Increasing at $x = a$
$f'(a) < 0$	$f'(a) = 0$	$f'(a) > 0$
		

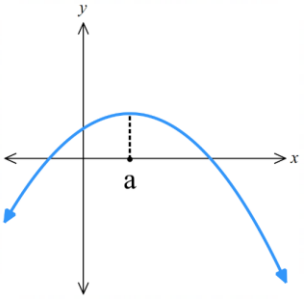
ⓘ **Identify** the behaviour of a curve at a point.

a



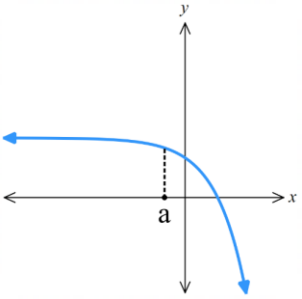
- ☐ Decreasing at $x = a$
- ☐ Stationary at $x = a$
- ☐ Increasing at $x = a$

b



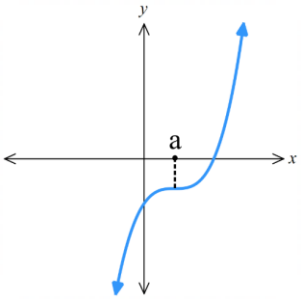
- ☐ Decreasing at $x = a$
- ☐ Stationary at $x = a$
- ☐ Increasing at $x = a$

c



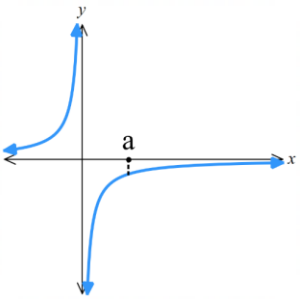
- ☐ Decreasing at $x = a$
- ☐ Stationary at $x = a$
- ☐ Increasing at $x = a$

d



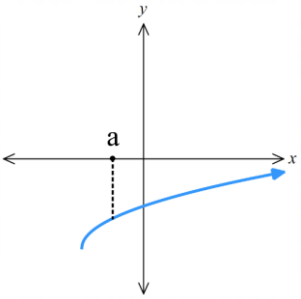
- ☐ Decreasing at $x = a$
- ☐ Stationary at $x = a$
- ☐ Increasing at $x = a$

e



- ☐ Decreasing at $x = a$
- ☐ Stationary at $x = a$
- ☐ Increasing at $x = a$

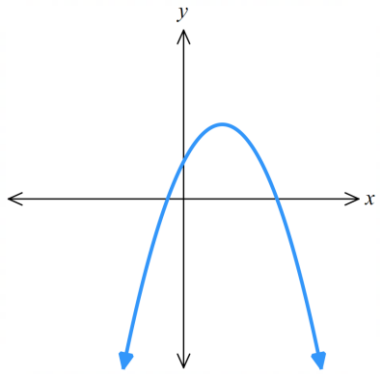
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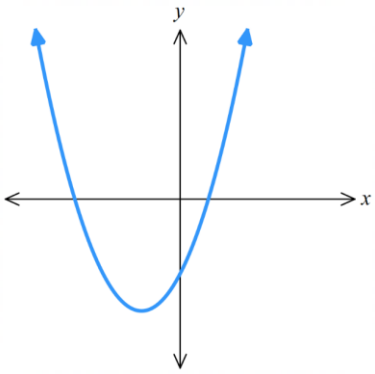
- ☐ Decreasing at $x = a$
- ☐ Stationary at $x = a$
- ☐ Increasing at $x = a$

Identify the sign of the derivative along a curve.

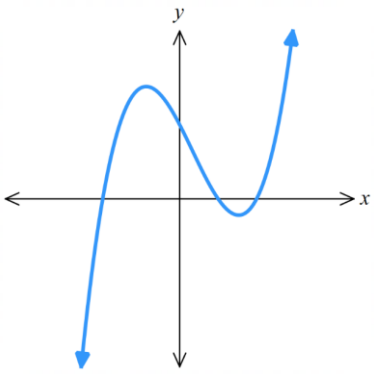
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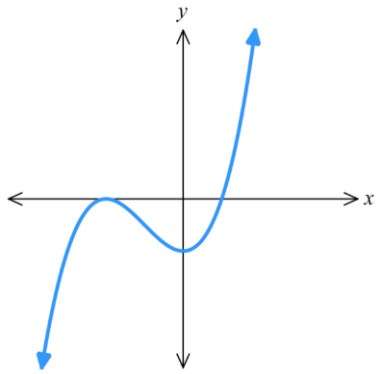
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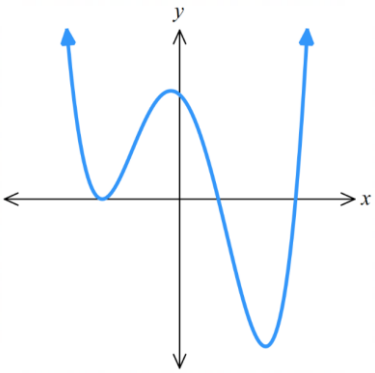
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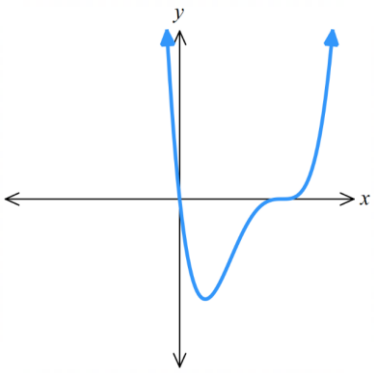
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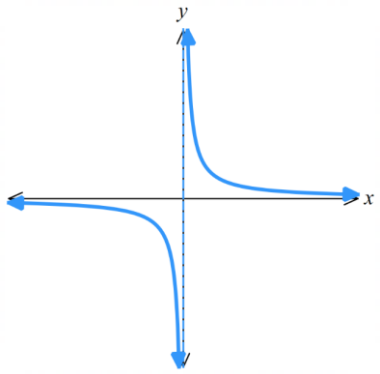
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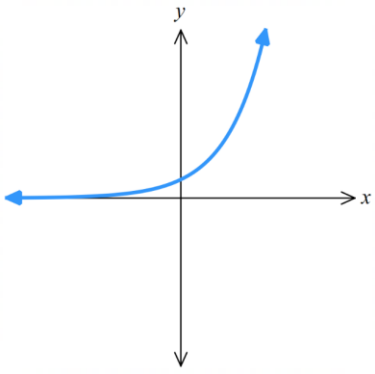
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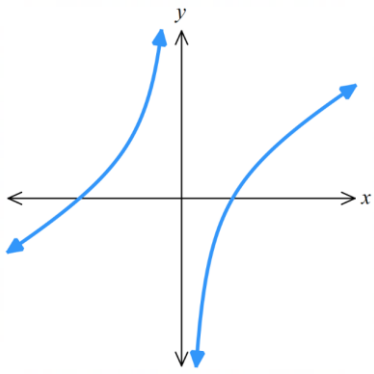
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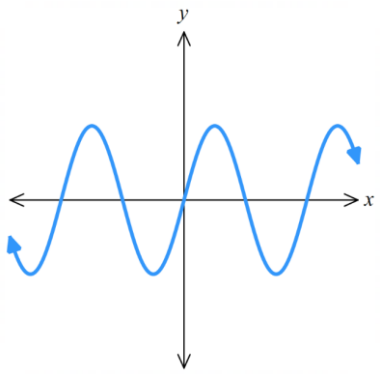
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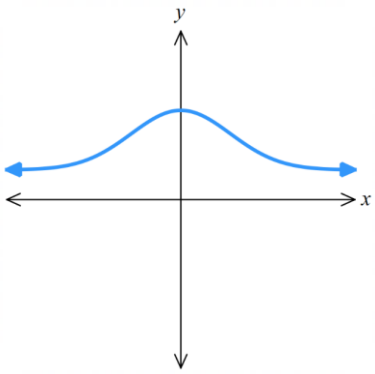
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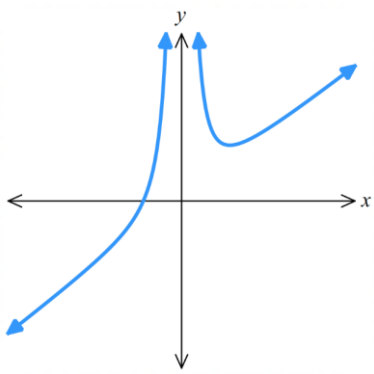
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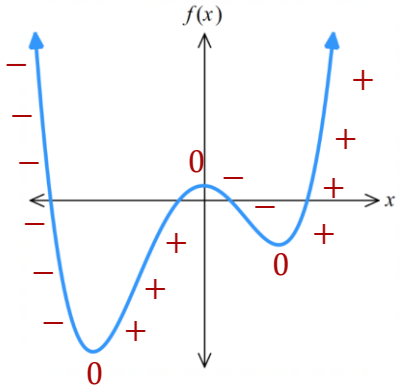
ⓘ Sketching the derivative from a given graph

1. Identify the sign of the gradient along $f(x)$.
2. Identify x values where $f(x)$ has horizontal tangents, and mark these as zeros of $f'(x)$.
3. Sketch $f'(x)$ using the sign of the gradient.

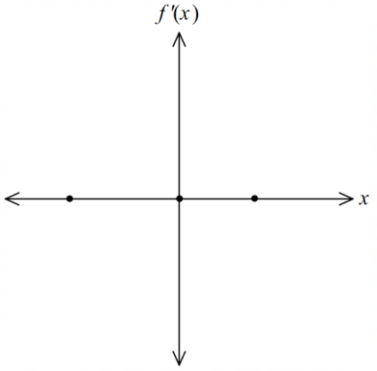
Example Sketch the derivative from a given graph.

Sketch the gradient function of the given graph of $f(x)$.

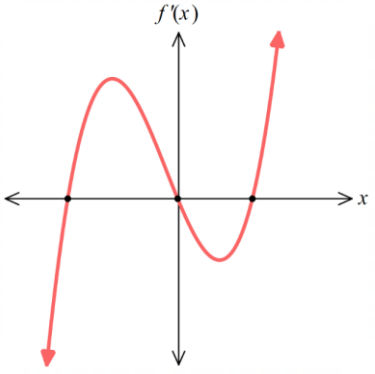
1. Sign of derivative



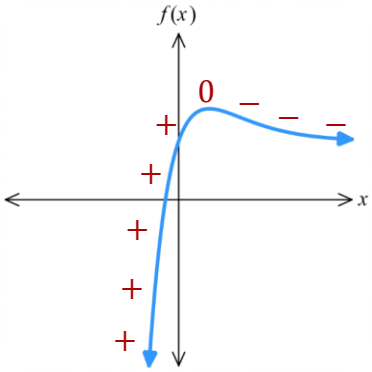
2. Identify $f'(x) = 0$



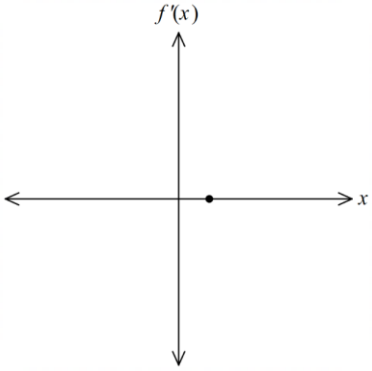
3. Sketch $f'(x)$



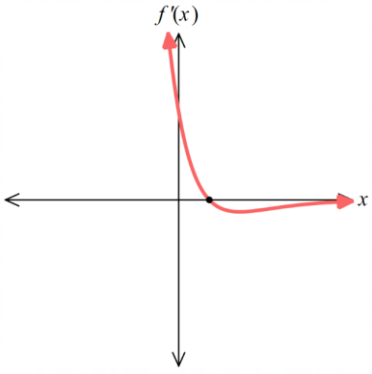
1. Sign of derivative



2. Identify $f'(x) = 0$



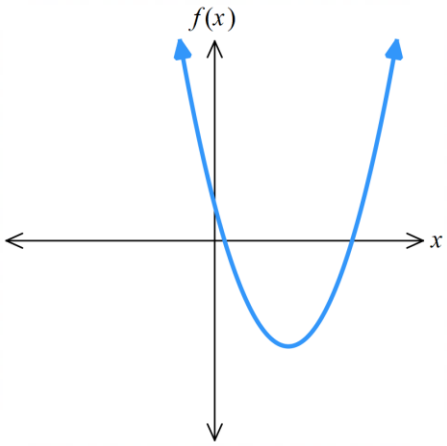
3. Sketch $f'(x)$

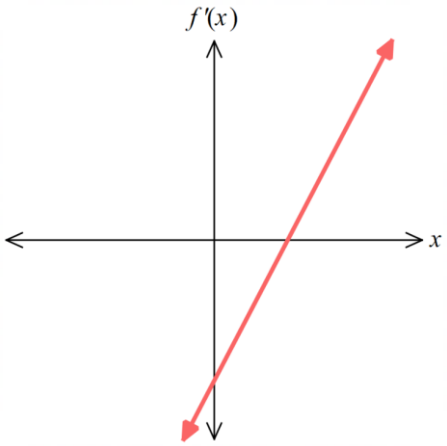


Practice

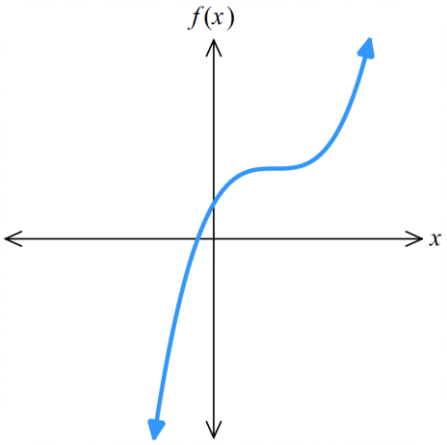
Sketch the derivative function

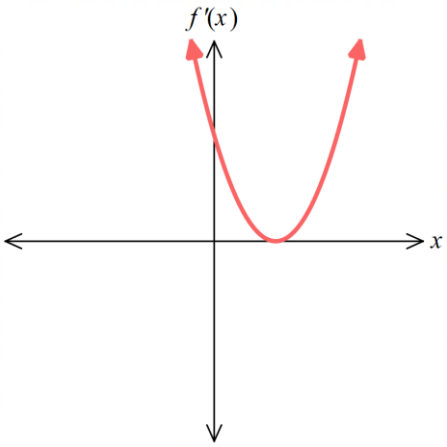
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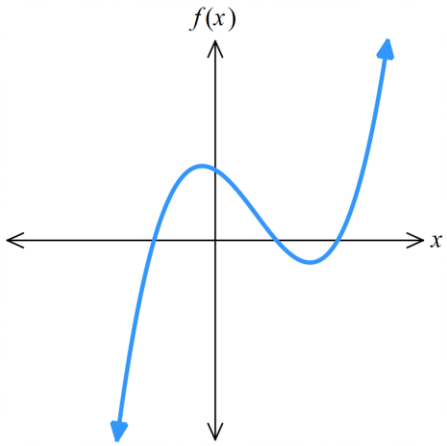


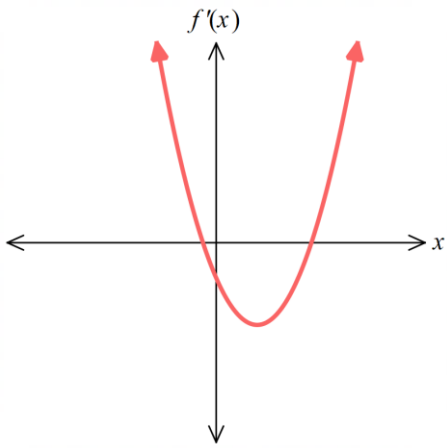
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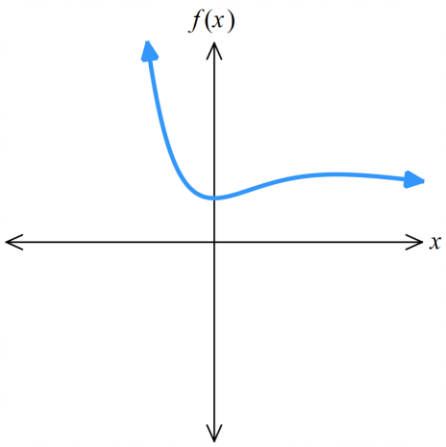


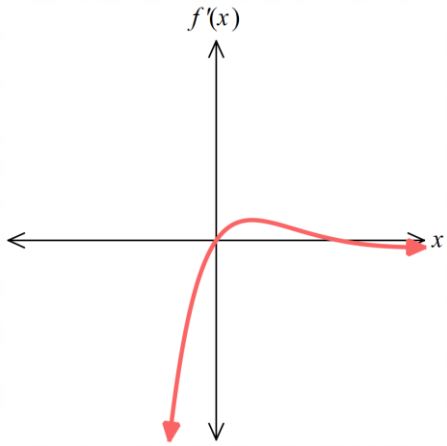
c





d





1 A function is increasing in a domain if the gradient of the tangent is p……

positive

2 A function is decreasing in a domain if the gradient of the tangent is n……

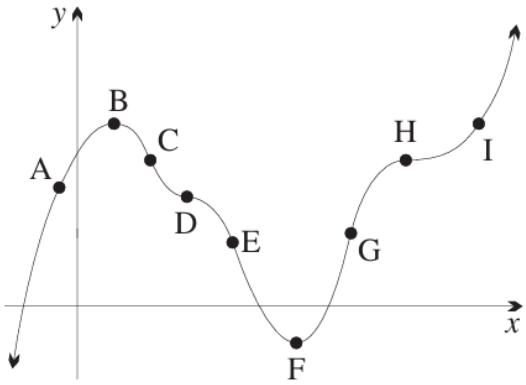
negative

3 When a function has a stationary point, the gradient of the tangent is ……

This can either be a t…… point or a point of i……

zero, turning, inflection

4 In the diagram, name the points where:



a $f'(x) > 0$.

A, G, I

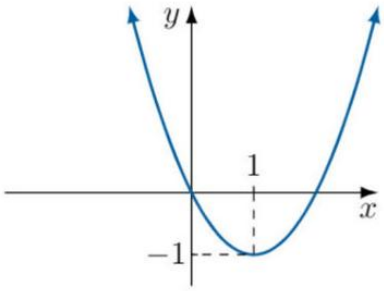
b $f'(x) < 0$.

C, E

c $f'(x) = 0$.

B, D, F, H

5 Consider the graph of $y = f(x)$



a State the domain where the function is increasing.

$x > 1$

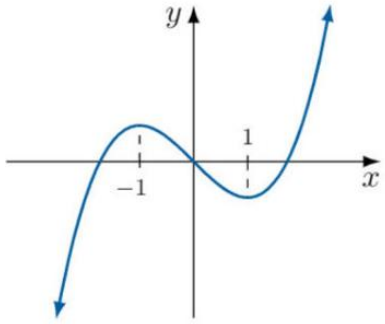
b State the domain where the function is decreasing.

$x < 1$

c State where the function has a horizontal tangent.

$x = 1$

6 Consider the graph of $y = f(x)$



a State where $f'(x) > 0$.

$x < -1$

$x > 1$

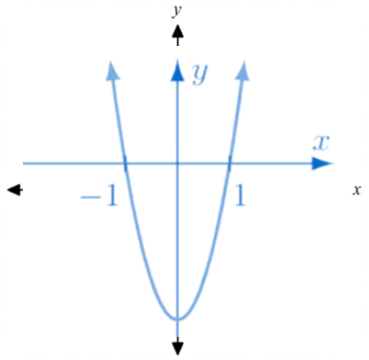
b State where $f'(x) < 0$.

$-1 < x < 1$

c State where $f'(x) = 0$.

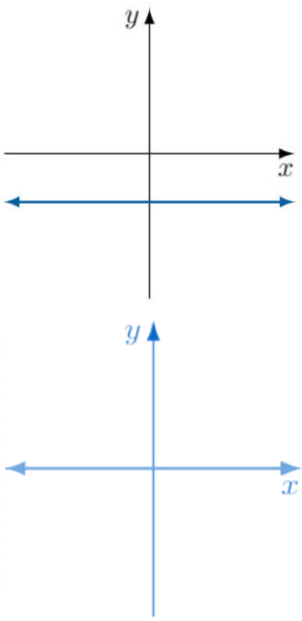
$x = -1, 1$

d Hence sketch $y = f'(x)$.

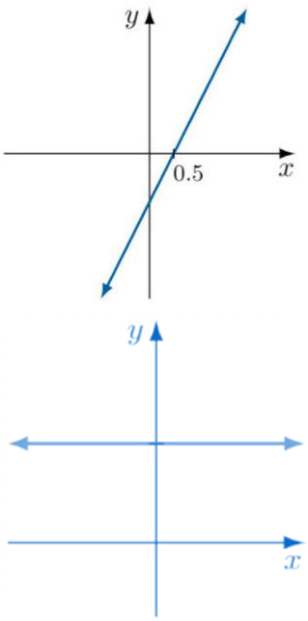


7 Sketch the derivative function for each curve.

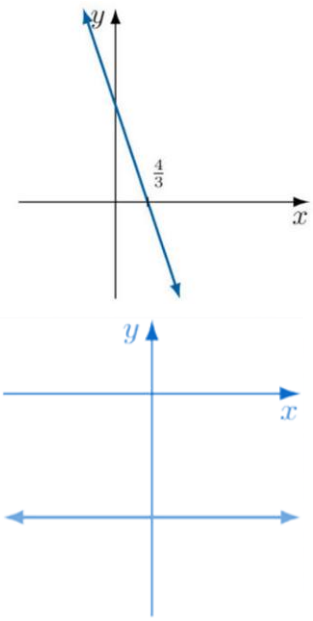
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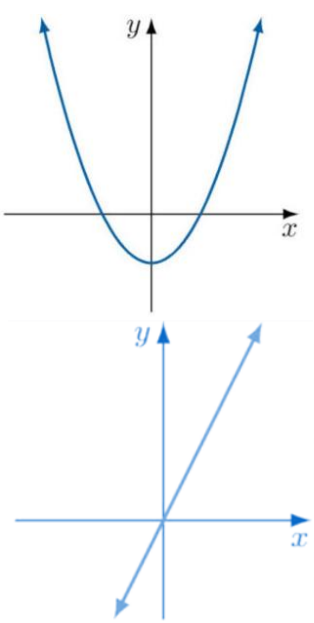
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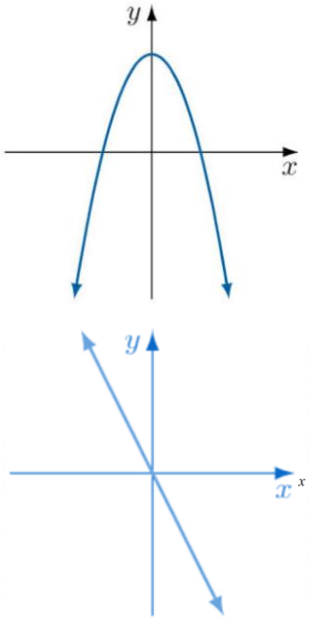
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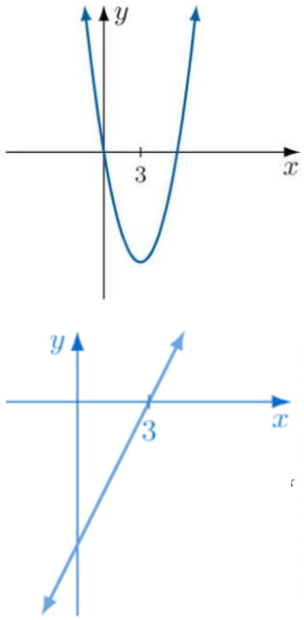
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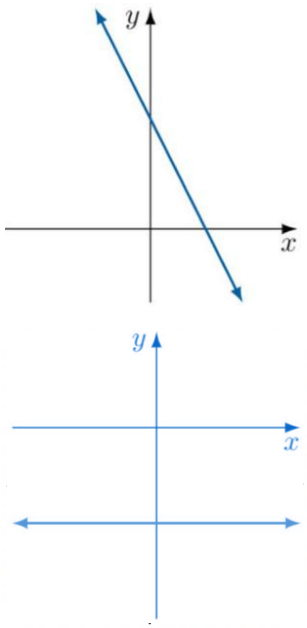
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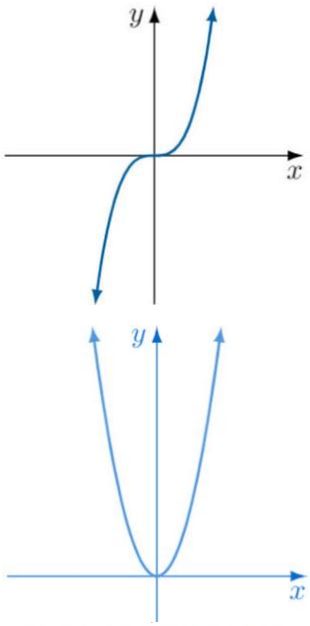
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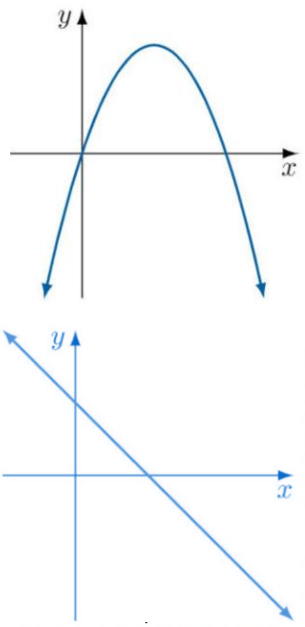
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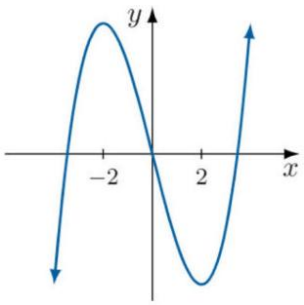
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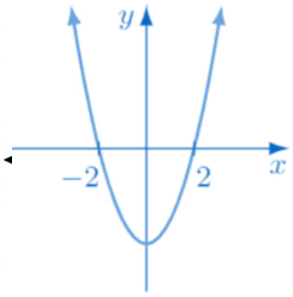


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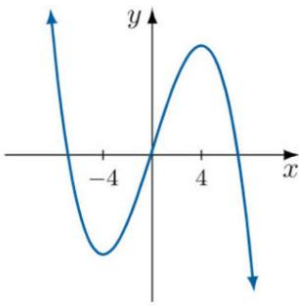


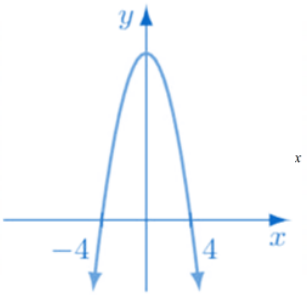
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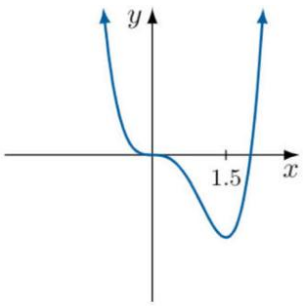


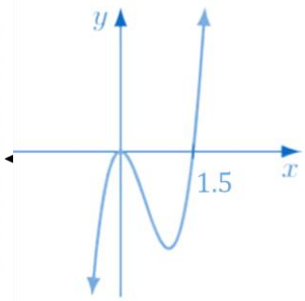
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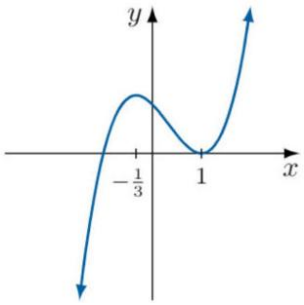


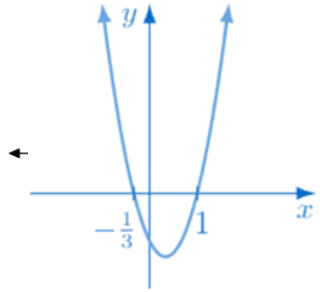
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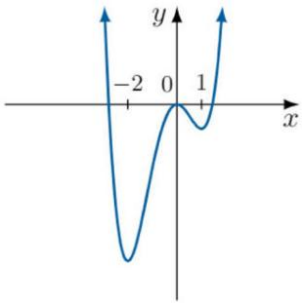


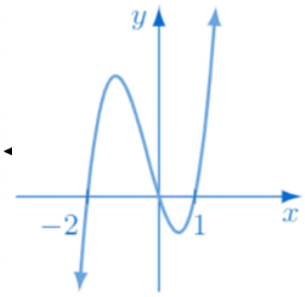
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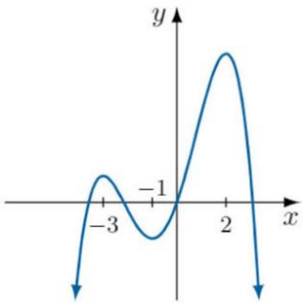


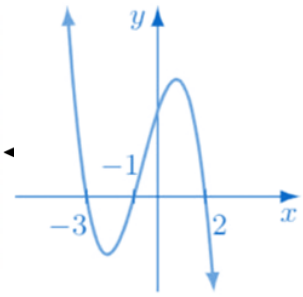
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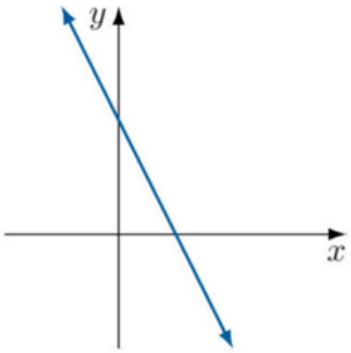


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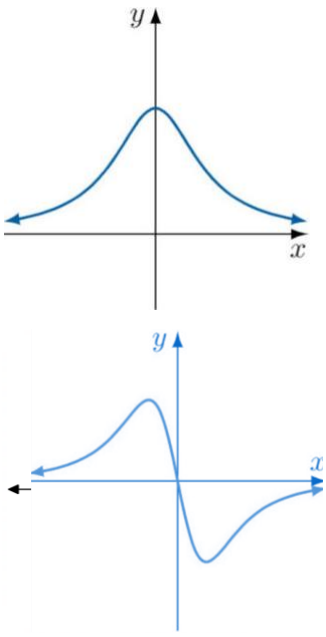
8 Describe the graph of $y = f(x)$ if the graph shown below is $y = f'(x)$.



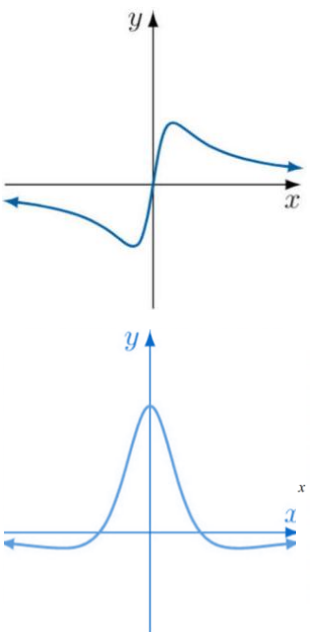
Concave down parabola with vertex to the right of the $y-$ axis.

9 Sketch the derivative function for each curve.

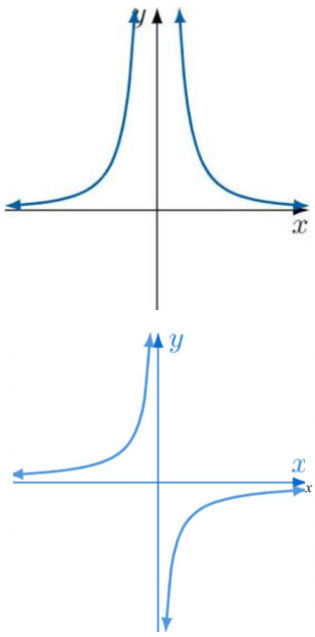
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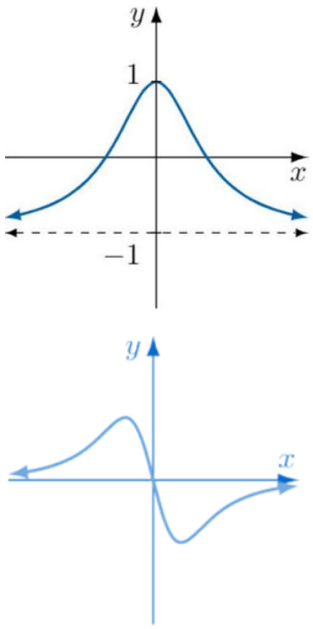
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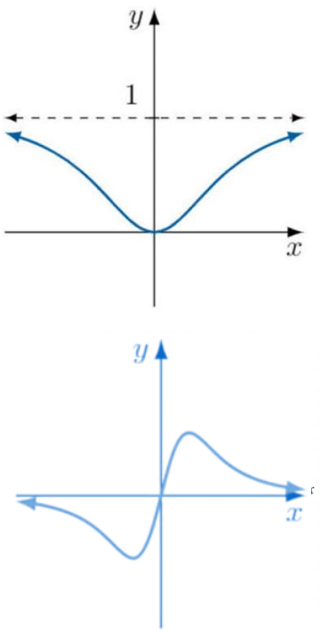
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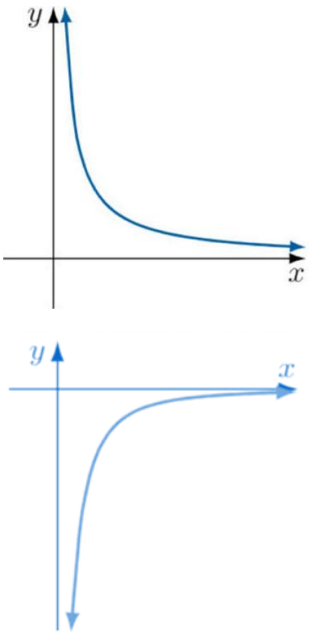
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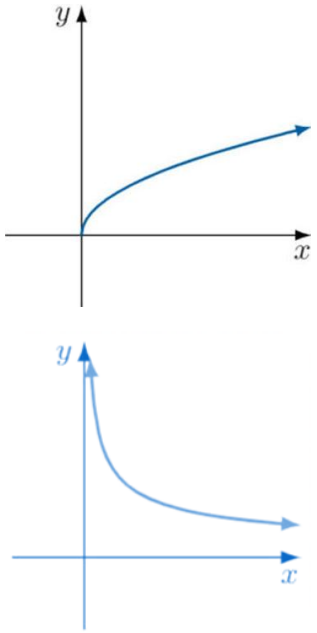
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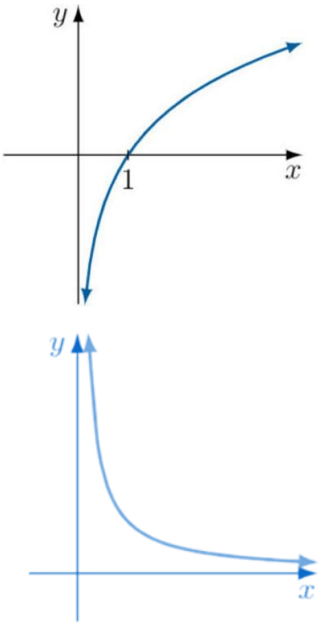
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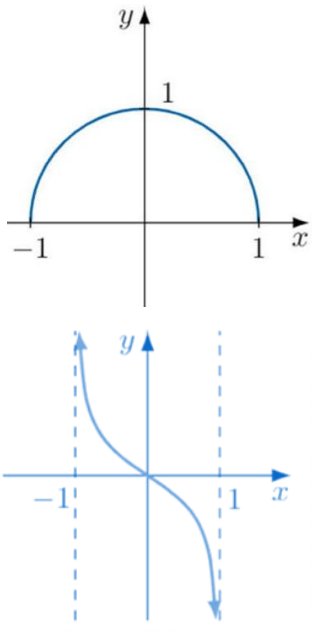
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h



i



Increasing Decreasing Stationary using Calculus

Review

☐ Solve inequalities

a $2x - 8 > 0$

b $8x + 9 < 0$

c $-2x + 3 > 0$

☐ Solve quadratic inequalities

a $(x - 2)(x + 5) < 0$

b $(2x - 1)(3x + 5) > 0$

c $x^2 - 5x + 6 < 0$

☐ Find the derivative at a point.

a $f(x) = x^3 - 2x^2 + 3.$
Find $f'(7)$

b $y = (x - 3)(x + 5).$
Find $f'(-2)$

c $f(x) = (x - 2)^3$
Find $f'(7)$

Determining the behaviour of a curve at a point

1. Differentiate to find $f'(x)$.
2. Evaluate $f'(a)$.
   a. Decreasing if $f'(a) < 0$.
   b. Increasing if $f'(a) > 0$.
   c. Stationary if $f'(a) = 0$.

Example Determine the behaviour of a curve at a point.

Differentiate $f(x) = x^3 - 12x$ and hence determine the behaviour of the curve at:

a $x = 5$

b $x = 2$

c $x = 0$

1. $f'(x) = 3x^2 - 12$

2.

a. $f'(5) = 63 > 0$

$\therefore$ Increasing at $x = 5$

b. $f'(2) = 0$

$\therefore$ Stationary at $x = 2$

c. $f'(0) = -12 < 0$

$\therefore$ Decreasing at $x = 0$

What is meant by "behaviour" of a function at a point?

Behaviour describes a function's .... at a point; it can be either increasing, decreasing, or stationary.

Why do we need to find the derivative first?

The derivative represents the ....

When is a curve increasing?

When $f'(x)$ .....

When is a curve decreasing?

When $f'(x)$ .....

When is a curve stationary?

When $f'(x)$ .....

Practice

$f(x) = x^3 + 2x^2 + 3$.

Determine the behaviour of the curve at

a $x = -2$

b $x = 0$

c $x = 2$

$f'(x) = 3x^2 + 4x$
$f'(-2) = 4 > 0$, increasing
$f'(0) = 0$, stationary
$f'(2) = 20 > 0$, increasing

$y = (x - 2)(x + 6)$.

Determine the behaviour of the curve at

a $x = -2$

b $x = 0$

c $x = 2$

$\frac{dy}{dx} = 2x + 4$
$\frac{dy}{dx}|_{-2} = 0$, stationary
$\frac{dy}{dx}|_{0} = 4$, increasing
$\frac{dy}{dx}|_{2} = 8 > 0$, increasing

Finding stationary points

1. Differentiate to find $f'(x)$.
2. Solve $f'(x) = 0$.
3. Substitute value of x from step 2 into $f(x)$ to find y-coordinate.

Example Find stationary points of a curve.

Find the stationary points of $f(x) = x^4 - 4x$.

1. $f'(x) = 4x^3 - 4$

2. Let $f'(x) = 0$

$4x^3 - 4 = 0$

$4x^3 = 4$

$x = 1$

3. $f(1) = (1)^4 - 4 = -3$

$\therefore$ Stationary point at $(1, -3)$

Find the stationary points of $f(x) = 3x^3 - 4x$.

1. $f'(x) = 9x^2 - 4$

2. Let $f'(x) = 0$

$9x^2 - 4 = 0$

$x = \pm\frac{2}{3}$

3. $f\left(-\frac{2}{3}\right) = \frac{16}{9}, f\left(\frac{2}{3}\right) = -\frac{16}{9}$

$\therefore$ Stationary points at $\left(-\frac{2}{3}, \frac{16}{9}\right), \left(\frac{2}{3}, -\frac{16}{9}\right)$

Practice

Find the stationary points of these functions. Check using Desmos.

a $f(x) = x^2 + 8x + 16$

$(-4, 0)$

b $f(x) = x^3 - 48x - 7$

$(-4, 121), (4, -135)$

c $f(x) = (x - 2)^4$

(2, 0)

d $f(x) = \frac{x-1}{x^2}$

$\left(2, \frac{1}{4}\right)$

Finding intervals where a curve is increasing or decreasing

1. Differentiate to find $f'(x)$
2. Solve to find:
   a. Increasing region $f'(x) > 0$.
   b. Decreasing region $f'(x) < 0$.

Example Find intervals where a curve is increasing or decreasing.

Determine the intervals where
$y = (x-2)(x-4)$
is increasing or decreasing.

1. $y = x^2 - 6x + 8$

$y' = 2x - 6$

2.

a. Increasing:

$2x - 6 > 0$

$x > 3$

b. Decreasing:

$2x - 6 < 0$

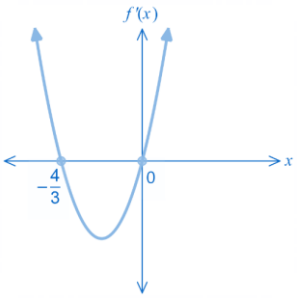
$x < 3$

Determine the intervals where
$f(x) = x^3 + 2x^2 + 3$
is increasing or decreasing.

1. $f'(x) = 3x^2 + 4x$

$= x(3x + 4)$

2.



a. Increasing:

$x(3x + 4) > 0$

$x > 0, x < -\frac{4}{3}$

b. Decreasing:

$x(3x + 4) < 0$

$-\frac{4}{3} < x < 0$

Practice

Find where the curve is increasing or decreasing.

a $y = -2x^2 + 8x - 1$

Increasing: $x < 2$
Decreasing: $x > 2$

b $y = 2x^2 - x$

Increasing: $x > \frac{1}{4}$
Decreasing: $x < \frac{1}{4}$

c $y = x^3 - 3x$

Increasing: $x < -1, x > 1$
Decreasing: $-1 < x < 1$

d $y = 4x^3 - 24x^2 + 36x - 8$

Increasing: $x < 1$ or $x > 3$
Decreasing: $1 < x < 3$

1

a Find the derivative of $f(x) = x^2 - 6x + 11$.

$f'(x) = 2x - 6$

b Hence determine whether the curve is increasing, decreasing, or stationary at:

$x = 0$	$x = 1$	$x = 3$	$x = 4$	$x = -1$
f'(0) = -6 (decreasing)	f'(1) = -4 (decreasing)	f'(3) = 0 (stationary)	f'(4) = 2 (increasing)	f'(-1) = -8 (decreasing)

2

a Find the derivative of $f(x) = x^3 - 6x^2 + 9$.

$f'(x) = 3x^2 - 12x$

b Hence determine whether the curve is increasing, decreasing, or stationary at:

$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = -1$
f'(0) = 0 (stationary)	f'(1) = -9 (decreasing)	f'(2) = -12 (decreasing)	f'(3) = -9 (decreasing)	f'(-1) = 15 (increasing)

3 Differentiate each function then determine whether the curve is increasing, decreasing, or stationary at $x = 0$.

a $y = (x - 1)^3$

$y' = 3(x - 1)^2$
y'(0) = 3(1)² = 3 (increasing)

b $y = (2x - 1)^4$

$y' = 8(2x - 1)^3$
y'(0) = 8(-1)³ = -8 (decreasing)

c $y = (x^2 + 3)^2$

$y' = 4x(x^2 + 3)$
y'(0) = 0 (stationary)

4 Differentiate each function then determine whether the curve is increasing, decreasing, or stationary at $x = 1$.

a $y = (x - 5)(x + 3)$

$y' = 2x - 2$
y'(1) = 0 (stationary)

b $y = (x - 2)(x^2 + 5)$

$y' = 3x^2 - 4x + 5$
y'(1) = 4 (increasing)

c $y = (x^4 + 2)(1 - x^3)$

$y' = 4x^3 - 7x^6 - 6x^2$
y'(1) = -9 (decreasing)

5 Differentiate each function then determine whether the curve is increasing, decreasing, or stationary at $x = 2$.

a $y = \frac{x}{x+1}$

$y' = \frac{1}{(x+1)^2}$

y'(2) = 1/9 (increasing)

b $y = \frac{x+1}{x-1}$

$y' = -\frac{2}{(x-1)^2}$

y'(2) = -2 (decreasing)

c $y = \frac{x^2}{x+2}$

$y' = \frac{x^2+4x}{(x+2)^2}$

$y'(2) = \frac{3}{4}$ (increasing)

6 Find the coordinates of the stationary points of each function.

a $y = x^2 - 2x$

(1, -1)

b $y = x^2 - 4x + 3$

(2, -1)

c $y = x^2 + 6x + 9$

(-3, 0)

d $y = 2x^2 - 16x$

(4, -32)

e $y = x^3 - 3x^2$

(0, 0), (2, -4)

f $y = x^3 - 12x$

(-2, 16), (2, -16)

7 Show that $y = -5x + 2$ is decreasing for all x.

$y' = -5 < 0$ for all x

8 Show that $y = x + 7$ is increasing for all x.

$y' = 1 > 0$ for all x

9 Show that $y = x^3$ is increasing for all values of x, apart from $x = 0$.

$y' = 3x^2 > 0$ for all x, except $x = 0$ where $y' = 0$

10 Show that $y = 3x^2$ is increasing for $x > 0$ and decreasing for $x < 0$. What happens at $x = 0$?

$y' = 6x$. When $x > 0, y' > 0$, when $x < 0, y' < 0$

when $x = 0, y' = 0$, so it is a stationary point.

11 Show that $y = \sqrt{x}$ is increasing for its entire domain.

$y' = \frac{1}{2\sqrt{x}}, y' > 0$ in the domain $x > 0$.

Note: this is not a rigorous enough proof, as the function is defined at $x = 0$ but the derivative is not, but this is sufficient for the HSC definition of increasing/decreasing functions.

12 $f(x) = 4x - x^2$

a Find $f'(x)$.

$f'(x) = 4 - 2x$

b For what values of x is $f(x)$:

i Increasing

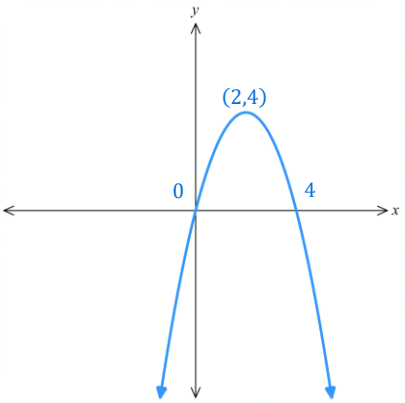
$x < 2$

ii Decreasing

$x > 2$

iii Stationary

$x = 2$



c By finding $f(2)$, label the sketch of $y = f(x)$ to the right.

13 $f(x) = x^3 - 3x^2 + 5$

a Find $f'(x)$

$f'(x) = 3x^2 - 6x$

b For what values of x is $f(x)$:

i Increasing

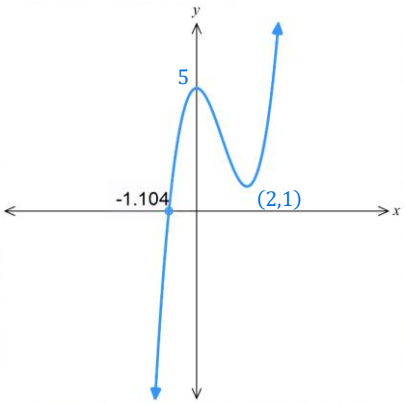
$x < 0, x > 2$

ii Decreasing

$0 < x < 2$

iii Stationary

$x = 0,2$



c By finding $f(0)$ and $f(2)$, label the sketch of $y = f(x)$ to the right.

14

a Differentiate $f(x) = -\frac{3}{x}$ and show that $f'(x) > 0$ for all x in its domain.

$f'(x) = \frac{3}{x^2} > 0$ for all x

b Explain why $f(-1) > f(2)$, despite $f'(x) > 0$.

There is an asymptote at $x = 0$. The hyperbola is not continuous.

15 Find the derivative of each function, and by solving $y' > 0$, find the values of x for which the function is increasing.

a $y = x^2 - 4x + 1$

$x > 2$

b $y = 7 - 6x - x^2$

$x < -3$

c $y = 2x^3 - 6x$

$x > 1$ or $x < -1$

d $y = x^3 - 3x^2 + 7$

$x < 0$ or $x > 2$

16 Find the values of x for which:

a $y = x^3 - x^2 - x - 1$ is decreasing.

$-\frac{1}{3} < x < 1$

b $y = x^3 - 3x^2 - 24x + 15$ is increasing.

$x < -2$ or $x > 4$

17 Use the quotient rule to find $f'(x)$ and hence show that:

a $f(x) = \frac{2x}{x-3}$ is decreasing for all x, $x \neq 3$.

$f'(x) = -\frac{6}{(x-3)^2} < 0$ for $x \neq 3$

b $f(x) = \frac{x^3}{x^2+1}$ is increasing for all x, except $x = 0$ where it is stationary.

$f'(x) = \frac{x^2(x^2+3)}{(x^2+1)^2} > 0$ for $x \neq 0$

18 **2021 HSC Advanced Band 4**

For what values of x is $f(x) = x^2 - 2x^3$ increasing?

$f'(x) = 2x - 6x^2 > 0$

$0 < x < \frac{1}{3}$

19 2010 HSC Advanced Band 5

$f(x) = x^3 - 3x^2 + kx + 8$. Find the values of k for which $f(x)$ is an increasing function.

$f'(x) = 3x^2 - 6x + k$

For $3x^2 - 6x + k > 0$:

Discriminant < 0 ($f'(x)$ has no x-intercepts.

$(-6)^2 - 4(3)(k) < 0$

$36 - 12k < 0$

$36 < 12k$

$k > 3$

https://www.desmos.com/calculator/cqn0cgsbj3

20 2018 HSC Advanced Band 5

$f(x) = x^3 + kx^2 + 3x - 5$. Find the values of k for which $f(x)$ has no stationary points.

$f'(x) = 3x^2 + 2kx + 3$

For no stationary points: $f'(x) \neq 0 \; for \; all \; x$

For quadratic $3x^2 + 2kx + 3$ to have no real roots:

Discriminant < 0

$(2k)^2 - 4(3)(3) < 0$

$4k^2 - 36 < 0$

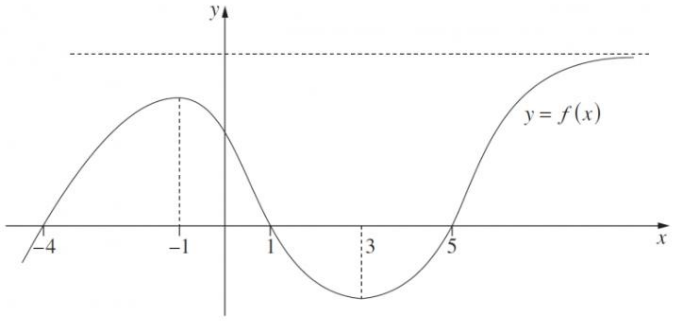
$(k-3)(k+3) < 0$

$-3 < k < 3$

https://www.desmos.com/calculator/b4rhzq7cea

21 2009 HSC Advanced Band 5

The diagram shows the graph of a function $y = f(x)$



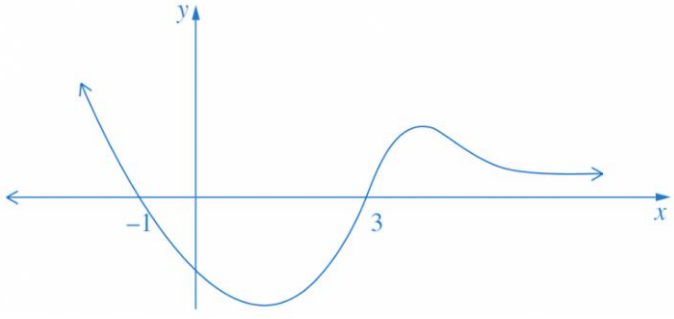
a For which values of x is the derivative $f'(x)$ negative?

$-1 < x < 3$

b What happens to $f'(x)$ for large values of x?

$f'(x) \to 0^+$

c Sketch the graph of $y = f'(x)$.



22

a Find $f'(x)$ for the function $f(x) = \frac{x^3}{3} + x^2 + 5x + 7$.

$f'(x) = x^2 + 2x + 5$

b By completing the square, show that $f'(x)$ is always positive, and hence that $f(x)$ is increasing for all x.

$f'(x) = (x + 1)^2 + 4,$

since $(x + 1)^2 \geq 0$, $f'(x) \geq 4 > 0$ so $f(x)$ is always increasing.

c Explain why the equation $f(x) = 0$ has exactly one real solution.

Since $f(x)$ is strictly increasing, is continuous for all x, and its domain is not restricted, it can cross the x-axis at most once.

Therefore, it has exactly one real solution.

23 For $f(x) = \frac{1-x^2}{x^2+1}$

a Find $f'(x)$.

b Evaluate $f'(0)$.

c Show that $f(x)$ is even.

d Hence explain why $f(x) \leq 1$ for all x.

a) $f'(x) = -\frac{4x}{(x^2+1)^2}$

b) $f'(0) = 0$

c) $f(-x) = \frac{1-(-x)^2}{(-x)^2+1}$

$= \frac{1-x^2}{x^2+1}$

$= f(x)$

Therefore f(x) is even.

d) we have $f'(x) < 0$ for $x > 0$

and $f'(x) > 0$ for $x < 0$

and $f'(x) = 0$ at $x = 0$,

therefore $f(x)$ has maximum value 1 at $x = 0$